

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027****Vision:** Dream of where you want.**Mission:** Means to achieve Vision**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

|      |                                 |                                                   |                                                                          |
|------|---------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|
| PEO1 | <b>Preparation</b>              | <b>P: Preparation</b>                             | <b>Pep-CL abbreviation<br/>pronounce as Pep-si-IL<br/>easy to recall</b> |
| PEO2 | <b>Core Competence</b>          | <b>E: Environment<br/>(Learning Environment)</b>  |                                                                          |
| PEO3 | <b>Breadth</b>                  | <b>P: Professionalism</b>                         |                                                                          |
| PEO4 | <b>Professionalism</b>          | <b>C: Core Competence</b>                         |                                                                          |
| PEO5 | <b>Learning<br/>Environment</b> | <b>L: Breadth (Learning in<br/>diverse areas)</b> |                                                                          |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)**Keywords of POs:**

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Sheikh Wasimuddin**  
Name of Student

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|                 |                      |                        |                                           |
|-----------------|----------------------|------------------------|-------------------------------------------|
| <b>Session</b>  | <b>2026-27 (ODD)</b> | <b>Course Name</b>     | <b>PE-V : Lab : Blockchain Technology</b> |
| <b>Semester</b> | <b>7</b>             | <b>Course Code</b>     | <b>23IOT1748</b>                          |
| <b>Roll No</b>  | <b>61</b>            | <b>Name of Student</b> | <b>Sheikh Wasimuddin</b>                  |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practical Number</b>   | <b>03</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Outcome</b>     | <ol style="list-style-type: none"> <li><b>1. Demonstrate basics of distributed systems and cryptographic primitives used in blockchain.</b></li> <li><b>2. Outline various research aspects for development of blockchain.</b></li> <li><b>3. Make use of different security protocols to build secure blockchain applications.</b></li> <li><b>4. Analyze smart contracts and Hyperledger Fabric used in blockchain.</b></li> <li><b>5. Determine the performance of applications developed with Bitcoin and Ethereum blockchain.</b></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Aim</b>                | Introduction to Ethereum, Installation and Setup of Remix IDE, and Exploration of Ethereum Accounts, Wallets, Gas, and Ether.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Theory (100 words)</b> | <p><b>1. Ethereum</b><br/>Ethereum is a blockchain network designed to support digital transactions as well as programmable applications.</p> <ul style="list-style-type: none"> <li>Unlike traditional applications, Ethereum-based programs can operate on a distributed network without requiring a single controlling organization.</li> <li>The platform uses <b>Ether (ETH)</b> as its primary digital currency.</li> <li>Developers can create <b>smart contracts</b> that automatically execute predefined rules when specified conditions are satisfied.</li> <li><b>Solidity</b> is one of the main programming languages used to develop smart contracts for Ethereum.</li> <li>Ethereum provides a distributed and tamper-resistant platform for developing applications such as decentralized finance, digital assets, and other blockchain solutions.</li> </ul> <p><b>2. Ethereum Accounts</b><br/>Ethereum accounts represent participants or programs that can communicate with the Ethereum network.</p> <ul style="list-style-type: none"> <li>An account is associated with a unique <b>Ethereum address</b>, which is used for identifying it during transactions.</li> <li>Ethereum primarily categorizes accounts into <b>Externally Owned</b></li> </ul> |



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**Accounts (EOAs) and Contract Accounts.**

- An EOA is operated by a person or application using a private key to authorize transactions.
- A Contract Account contains smart contract code and performs actions according to the instructions programmed into the contract.
- Accounts can receive and transfer ETH and can also participate in interactions with deployed smart contracts.

**3. Ethereum Wallets**

An Ethereum wallet is an application or physical device that provides users with a way to access and manage their blockchain accounts.

- It can be used to generate addresses, approve transactions, send ETH, receive assets, and communicate with decentralized applications.
- Wallets protect access to accounts through mechanisms such as **private keys, seed phrases, or recovery phrases**.
- Common categories include **software wallets, hardware wallets, and browser-integrated wallets**.
- The actual ETH balance is maintained on the Ethereum blockchain rather than being physically contained inside the wallet.
- A wallet essentially provides the tools and credentials needed to securely interact with blockchain-based assets and services.

**4. Gas**

Gas represents the amount of computational effort needed to execute an operation on the Ethereum network.

- Every transaction or smart contract operation consumes a specific quantity of gas depending on the work performed.
- Users pay for this computational resource through **gas fees**, which are normally paid in ETH.
- Operations involving more computation or complex smart contract logic generally consume more gas.
- The gas mechanism helps allocate network resources and discourages unnecessary or excessive use of computational capacity.
- Gas fees also compensate the network participants responsible for processing and validating transactions.

**5. Ether (ETH)**

Ether, commonly represented by **ETH**, is the main digital asset associated with the Ethereum network.

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|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <ul style="list-style-type: none"> <li>It can be transferred between Ethereum addresses as a form of digital value.</li> <li>ETH is also used to pay the costs associated with transactions and smart contract execution.</li> <li>The smallest denomination of Ether is called <b>wei</b>, where <b>1 ETH = <math>10^{18}</math> wei</b>.</li> <li>Users generally need ETH to perform transactions and interact with applications deployed on Ethereum.</li> <li>Apart from its use as a payment asset, ETH plays an important role in maintaining the operation and security of the Ethereum ecosystem.</li> </ul> <p><b>6. Remix IDE</b></p> <p><b>Remix IDE</b> is a browser-based development platform designed primarily for creating Ethereum smart contracts.</p> <ul style="list-style-type: none"> <li>It provides an environment in which developers can write and compile programs using <b>Solidity</b>.</li> <li>After compilation, a smart contract can be deployed and tested directly through the Remix interface.</li> <li>Remix provides testing environments such as <b>Remix VM</b>, allowing developers to experiment with contracts without spending real ETH.</li> <li>The platform also provides tools for examining contract functions, transactions, and execution results.</li> <li>Because it requires minimal setup and provides an integrated development environment, Remix is particularly useful for students and beginners learning Ethereum smart contract development.</li> </ul> |
| <p>Procedure and Execution</p> <p>(100 Words)</p> | <ol style="list-style-type: none"> <li>Launch a web browser and open the <b>Remix IDE</b> platform.</li> <li>Create a new Solidity source file and save it using the .sol file extension.</li> <li>Develop a basic smart contract by writing the required Solidity code.</li> <li>Choose a compatible <b>Solidity compiler version</b> and compile the source code to check for errors.</li> <li>Navigate to the <b>Deploy &amp; Run Transactions</b> panel available in Remix IDE.</li> <li>Choose <b>Remix VM</b> as the execution environment to test the contract locally without using real ETH.</li> <li>Deploy the compiled contract and examine the information generated for the deployment transaction.</li> <li>View the Ethereum account provided by the selected environment and check its address and available balance.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



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9. Examine how an Ethereum wallet provides access to accounts and protects credentials such as private keys.
10. Monitor the **gas consumption** associated with contract deployment and function execution.
11. Understand the role of **Ether (ETH)** in covering transaction costs and gas fees on the Ethereum network.
12. Execute the available functions of the deployed smart contract and note the output or transaction results.
13. Record the observations, transaction details, and obtained results in the practical record.

### Stepwise Screenshots with steps:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 contract HelloEthereum {
5     function printMessage() public pure returns (string memory) {
6         return "Hello this is sheikh wasimuddin";
7     }
8 }
```

COMPILER 0.8.34+commit.80d5c536

Auto compile  
Hide warnings  
Advanced Configurations

Compile lab3\_sheikhwasim...

Compile and Run script

CONTRACT: HelloEthereum (lab3\_sheikhwasir)

Run Remix Analysis  
Run SolidityScan  
Publish on IPFS  
Publish on Swarm  
Compilation Details



Nagar YuwakShikshanSanstha's  
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The screenshot displays the Remix IDE interface, which is used for developing and deploying smart contracts on the Ethereum blockchain. The interface is divided into several panels:

- Left Panel (Environment and Deploy):** Shows the 'Environment' section with 'Remix VM' selected and 'Osaka' as the network. The 'Account' is '0x5E1...ad5C4' with a balance of '99,999 ETH'. The 'Deploy' section shows the 'HelloEthereum' contract is 'Compiled'. The 'Value' is set to '0' and the 'Gas limit' is 'auto'. A 'Deploy' button is visible.
- Top Right Panel (Code Editor):** Displays the Solidity code for the 'HelloEthereum' contract. The code includes a pragma statement for Solidity version 0.8.20, a contract definition, and a 'printMessage' function that returns the string 'Hello this is sheikh wasimuddin'.
- Bottom Right Panel (Execution Console):** Shows the output of the contract execution. It indicates that the transaction was successful and provides details such as the transaction hash, block hash, block number, contract address, and the gas used (12479 gas).



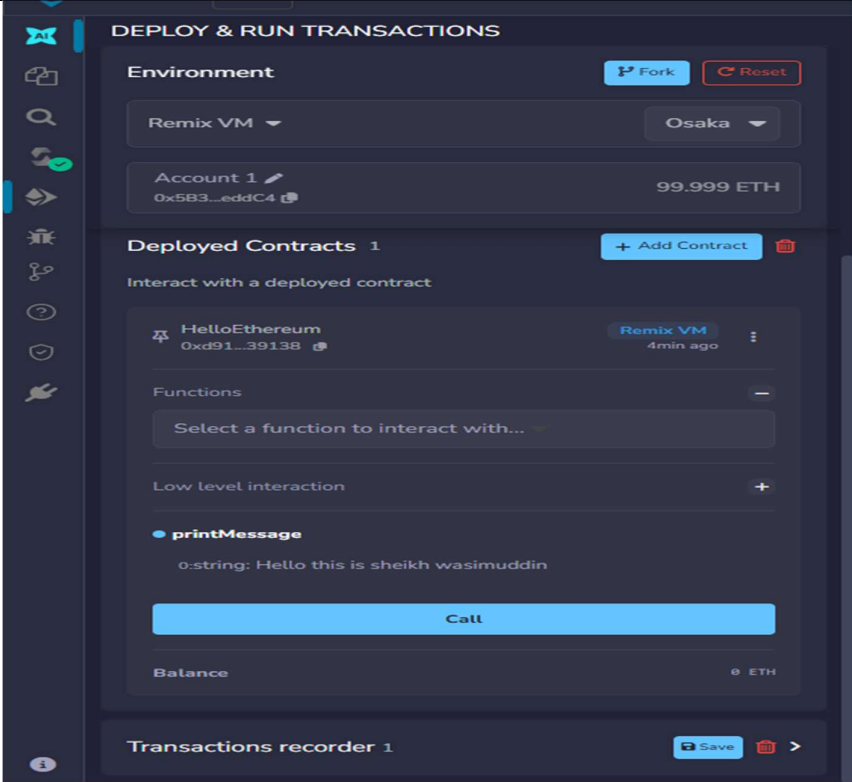
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|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Conclusion | <p>The practical was successfully completed using <b>Remix IDE</b> and the Solidity smart contract lab3_sheikhwasimuddin.sol. The HelloEthereum contract was compiled successfully and deployed on the <b>Remix VM (Osaka)</b> environment. The deployment transaction was completed successfully, and the deployed contract was accessed through the generated contract address. The printMessage() function was executed successfully and produced the output “Hello this is sheikh wasimuddin”. The practical also demonstrated the use of an Ethereum account, deployment transactions, gas consumption, and interaction with a deployed smart contract. Thus, the experiment provided practical knowledge of <b>Solidity programming, Ethereum smart contract deployment, and contract execution using Remix IDE.</b></p> |
| Date       | 11/08/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |